

Auditing Model Substitution in LLM APIs:

Why Software Tests Fail and Hardware Attestation Wins

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The Model Substitution Problem: Economic Incentives Drive Covert Swaps

- Commercial LLM APIs charge for specific models (e.g., Llama-3-70B, Qwen2-72B).
- Black-box APIs give users no guarantee the advertised model is actually served.
- Providers face strong incentives to substitute cheaper alternatives while maintaining pricing.
- Formalized as hypothesis testing: distinguish $P_{\text{spec}}(y|x)$ from $P_{\text{actual}}(y|x)$.

Four Substitution Types

1. Smaller models — serve 8B instead of 70B
2. Quantized variants — FP8 / INT8 instead of FP16
3. Fine-tuned or updated versions — silently patched weights
4. Entirely different model families — swap architecture altogether

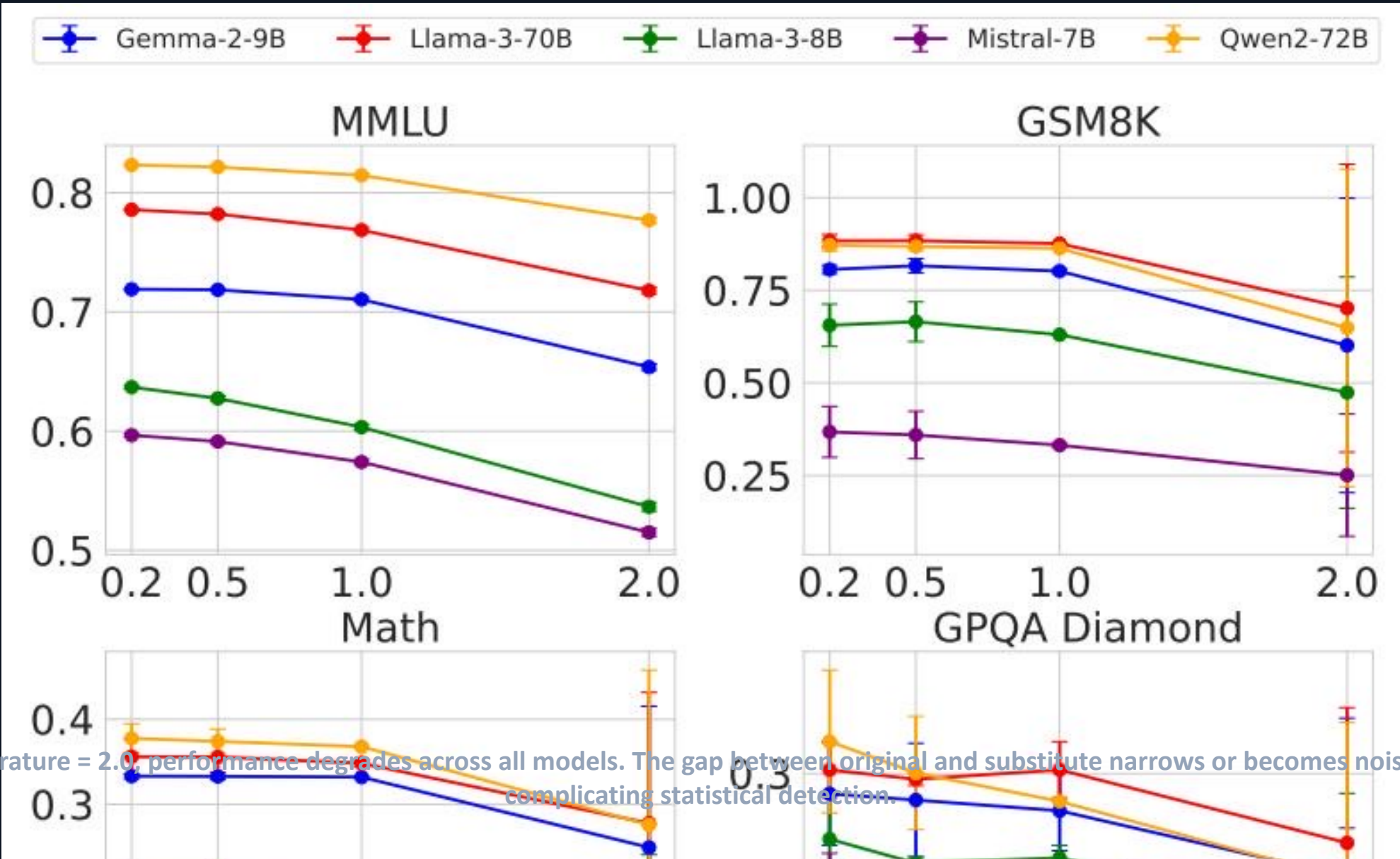
Core insight: Any substitution that preserves plausible outputs defeats naive black-box auditing.

FP8 Quantization Preserves Most Accuracy — Making Substitution Ha

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Observation: For most models, FP8 scores fall within the error bars of the original, especially on MMLU and GSM8K.

Higher Temperature Amplifies Performance Gaps but Increases Noise



Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same Llama-3-8B produces different log probabilities across transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2, and A100 vs H100.

An auditor cannot distinguish legitimate infrastructure differences from actual model substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-

- Provider routes only a fraction p of queries to the cheaper model.
- Mixed distribution: $(1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{substitute}}$
- As $p \rightarrow 0$, the mixture converges to P_{spec} .
- Required sample size grows as $O(1/p^2)$ for reliable detection.
- At $p = 0.05$, an auditor needs thousands of queries per prompt — economically infeasible.

Mixed Output Distribution

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$

When p is small, total-variation distance between P_{mix} and P_{spec} shrinks proportionally, making any statistical test query-prohibitive.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Core Weakness	Practical Limitation	Verdict
Text-based statistical tests (e.g., MMD)	Require impractically large query budgets; fail against subtle or randomized substitutions.	Sample complexity scales poorly with output diversity and mixing rate p .	Unreliable
Log-probability fingerprinting	Production inference is nondeterministic across software versions and GPU hardware.	Same model on A100 vs H100 or transformers 4.40 vs 4.50 yields different log-probs.	Unreliable
Adversarial countermeasures	Providers can detect audit prompts and route them to the genuine model (benchmark evasion).	Randomized routing further dilutes the substitution signal below detection threshold.	Unreliable

Conclusion: No software-only method provides a robust, practical guarantee against a motivated provider.

Trusted Execution Environments Provide Provable Cryptographic Mod

- Hardware-secured enclaves with cryptographic attestation of code and data.
- Provider proves that specific model weights are loaded.
- Auditor verifies that correct inference code is running inside the enclave.
- Only modest performance overhead vs. plaintext inference.
- Unlike ZKPs, TEEs are practical and deployable today for large models.

Practical Deployment Path

NVIDIA Confidential Computing on H100 GPUs

- GPU-isolated memory encryption
- Remote attestation via CPU TEE
- Verifiable boot chain for inference stack

TEEs shift the trust model from 'trust the provider' to 'trust the hardware attestation chain'.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and practically undetectable via software alone.
2. FP8 quantization barely affects benchmark scores — the signal is buried in noise.
3. Inference nondeterminism across frameworks and GPUs kills log-probability fingerprinting.
4. Randomized substitution at low mixing rates defeats statistical tests by diluting the signal.
5. Trusted Execution Environments are the only currently viable path to verifiable LLM APIs.

Call to action: Adopt hardware attestation standards for production LLM APIs. Demand TEE-backed inference from your cloud providers.